

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

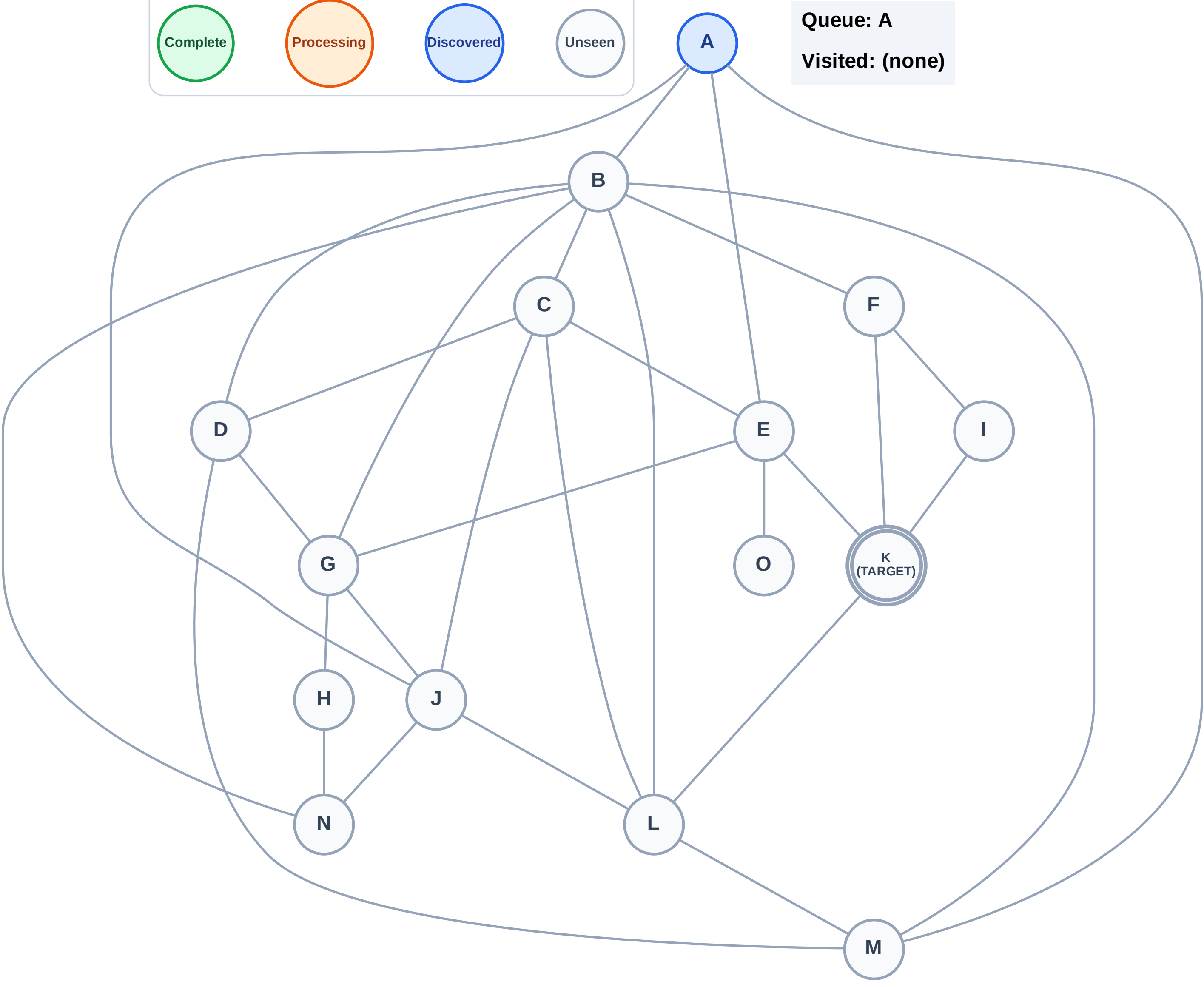
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

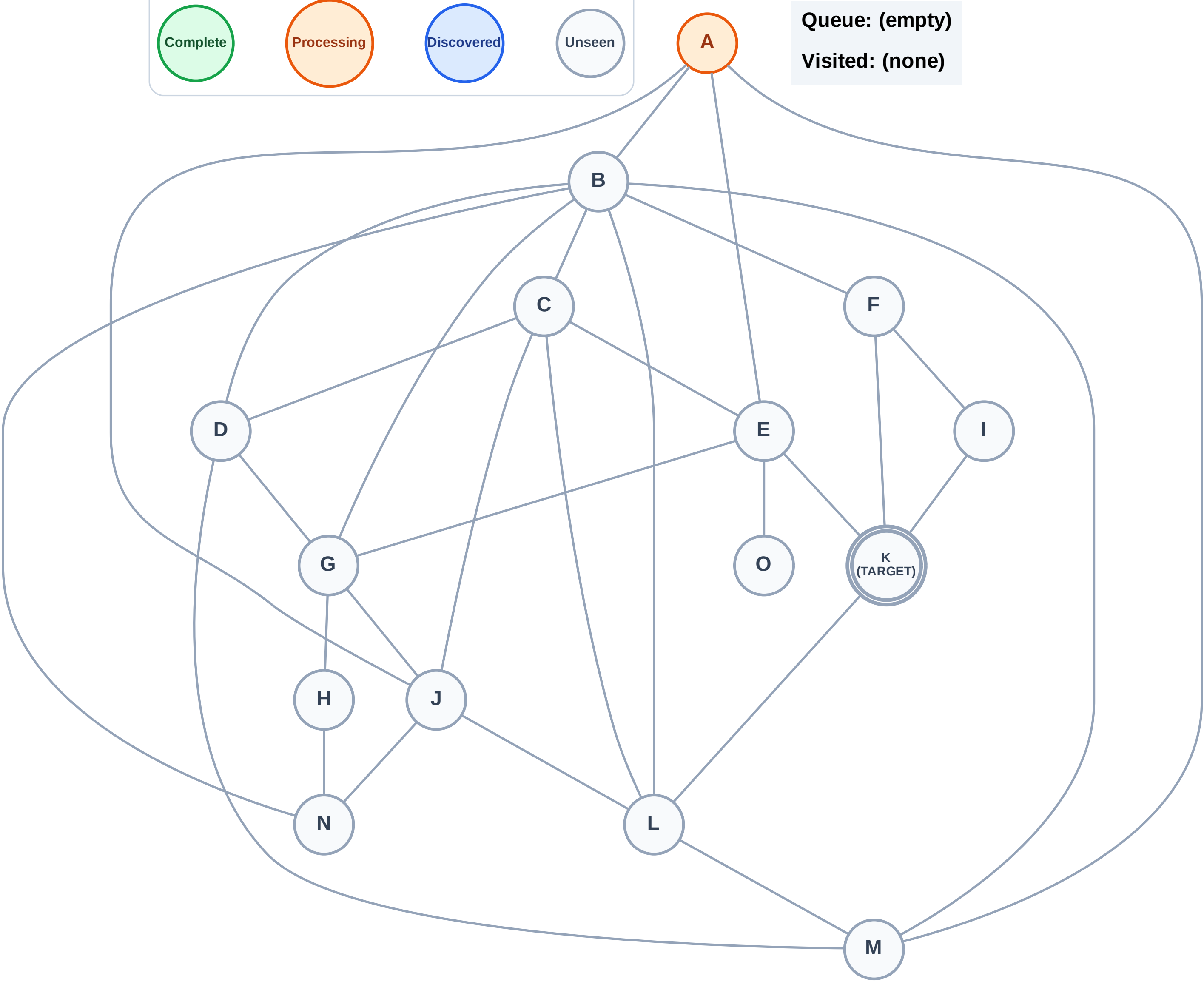
Complete

Processing

Discovered

Unseen

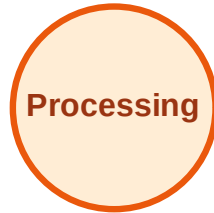
Queue: (empty)
Visited: (none)



BFS Traversal

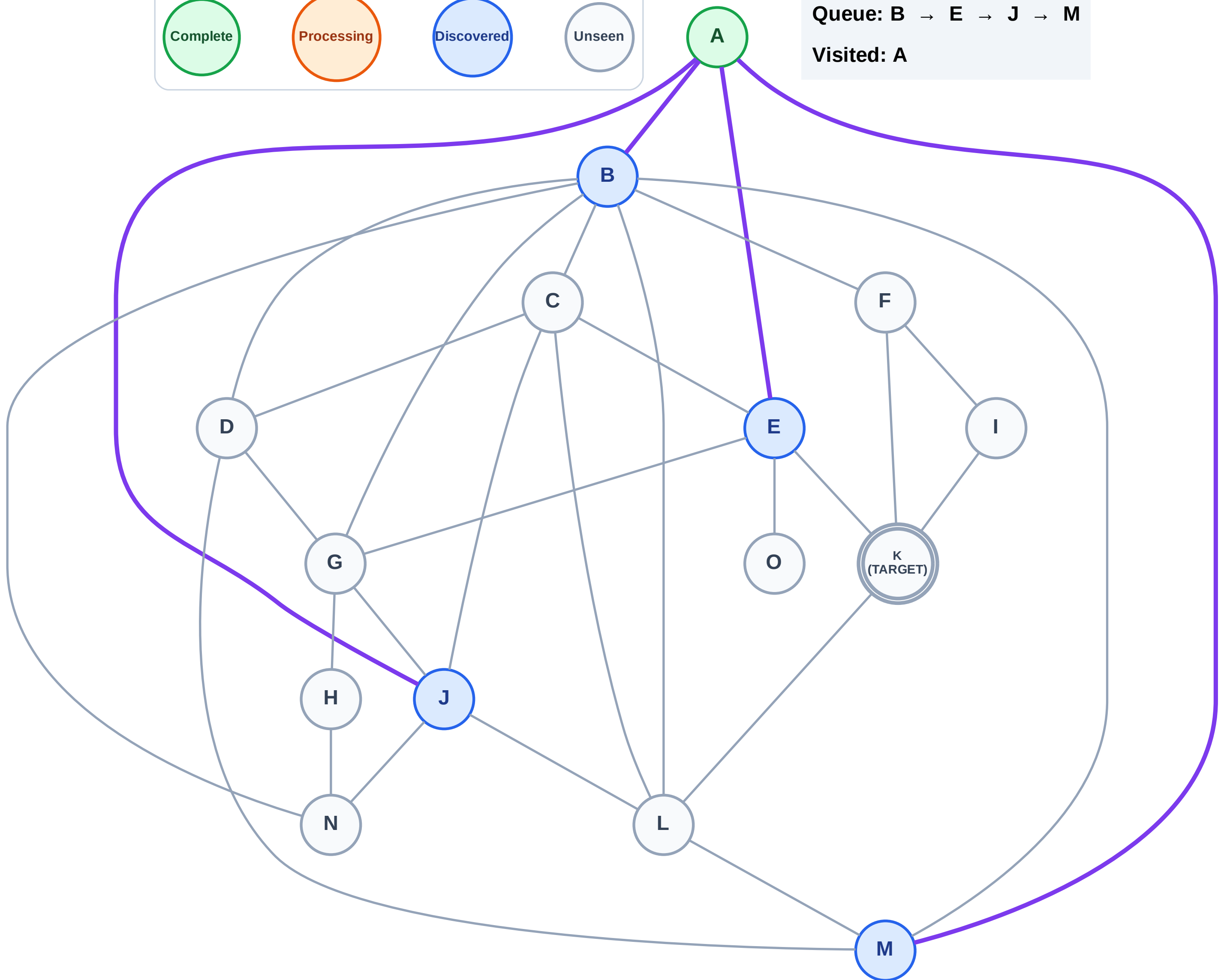
Step 3: Discover B, E, J, M from A

Legend



Queue: B → E → J → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

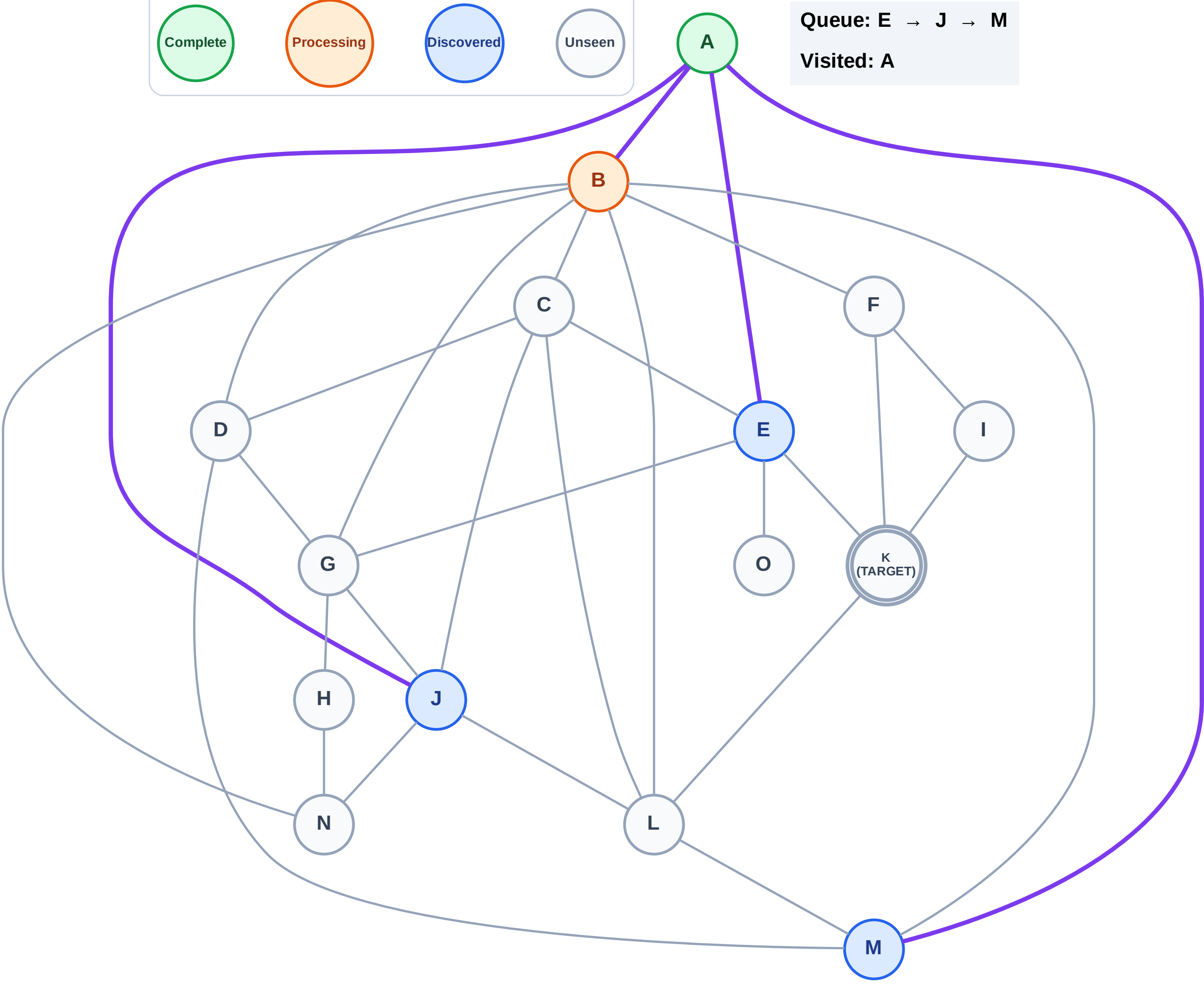
Processing

Discovered

Unseen

Queue: E → J → M

Visited: A



BFS Traversal

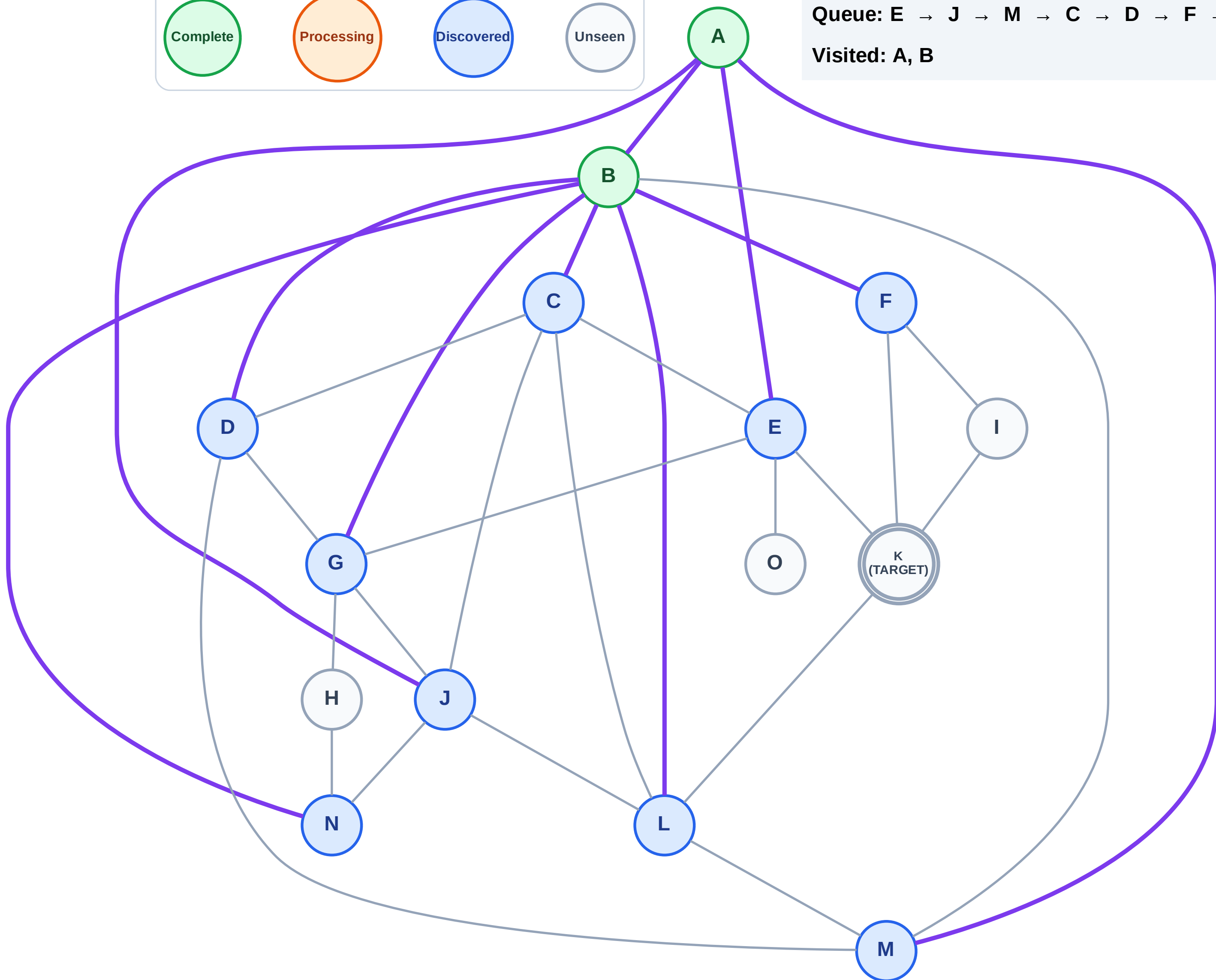
Step 5: Discover C, D, F, G, L, N from B

Legend



Queue: E → J → M → C → D → F → G → L → N

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

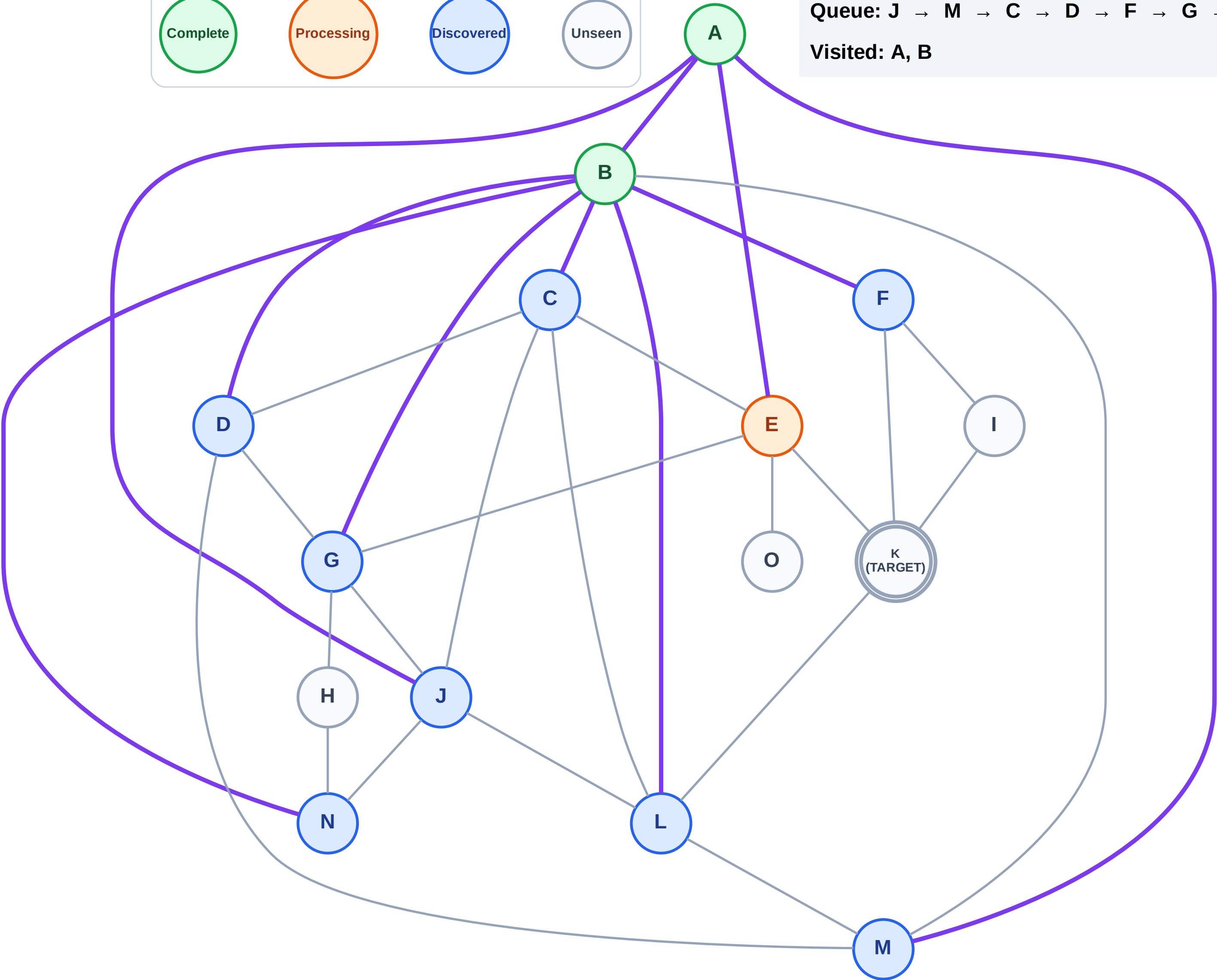
Processing

Discovered

Unseen

Queue: J → M → C → D → F → G → L → N

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend

Complete

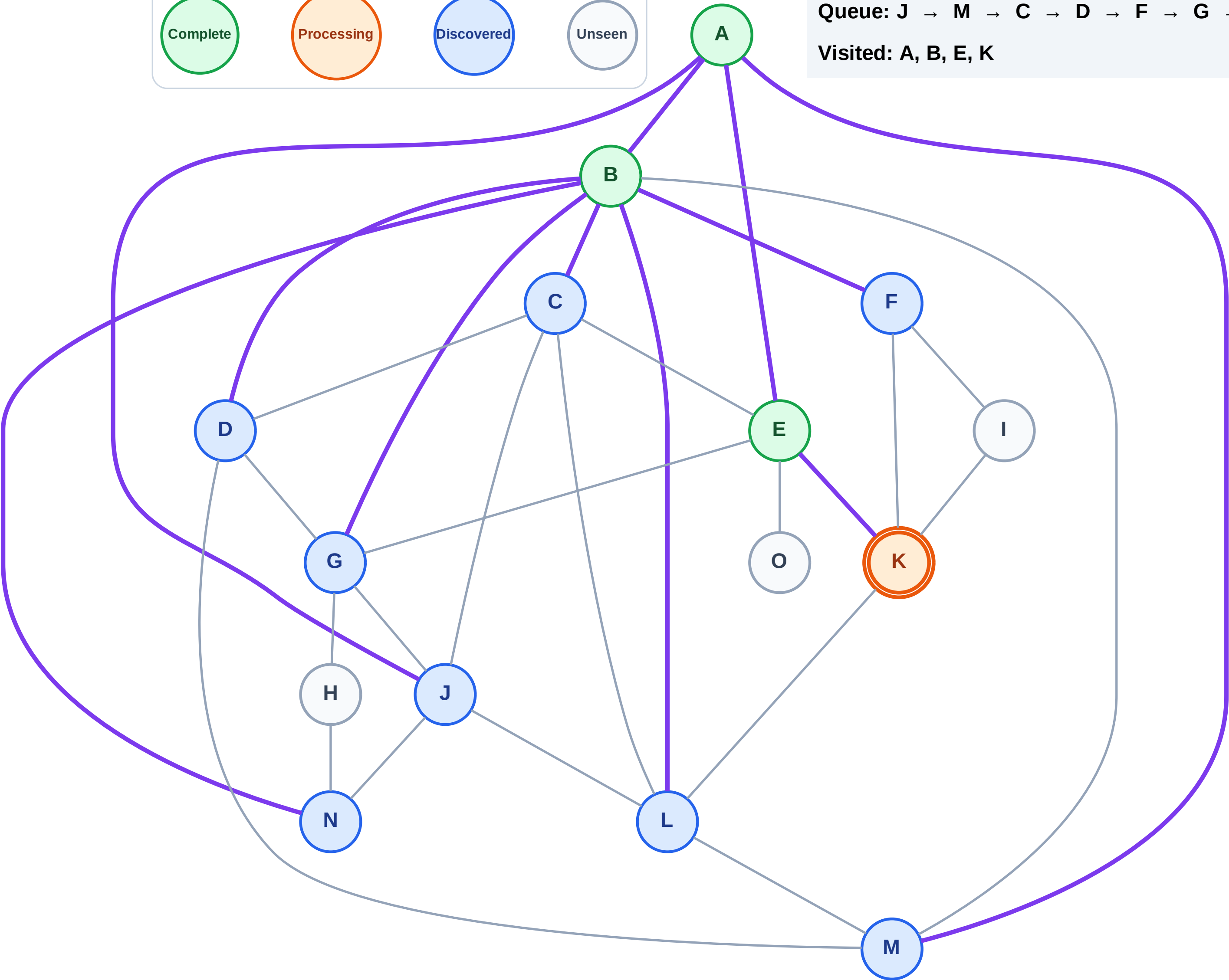
Processing

Discovered

Unseen

Queue: J → M → C → D → F → G → L → N

Visited: A, B, E, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

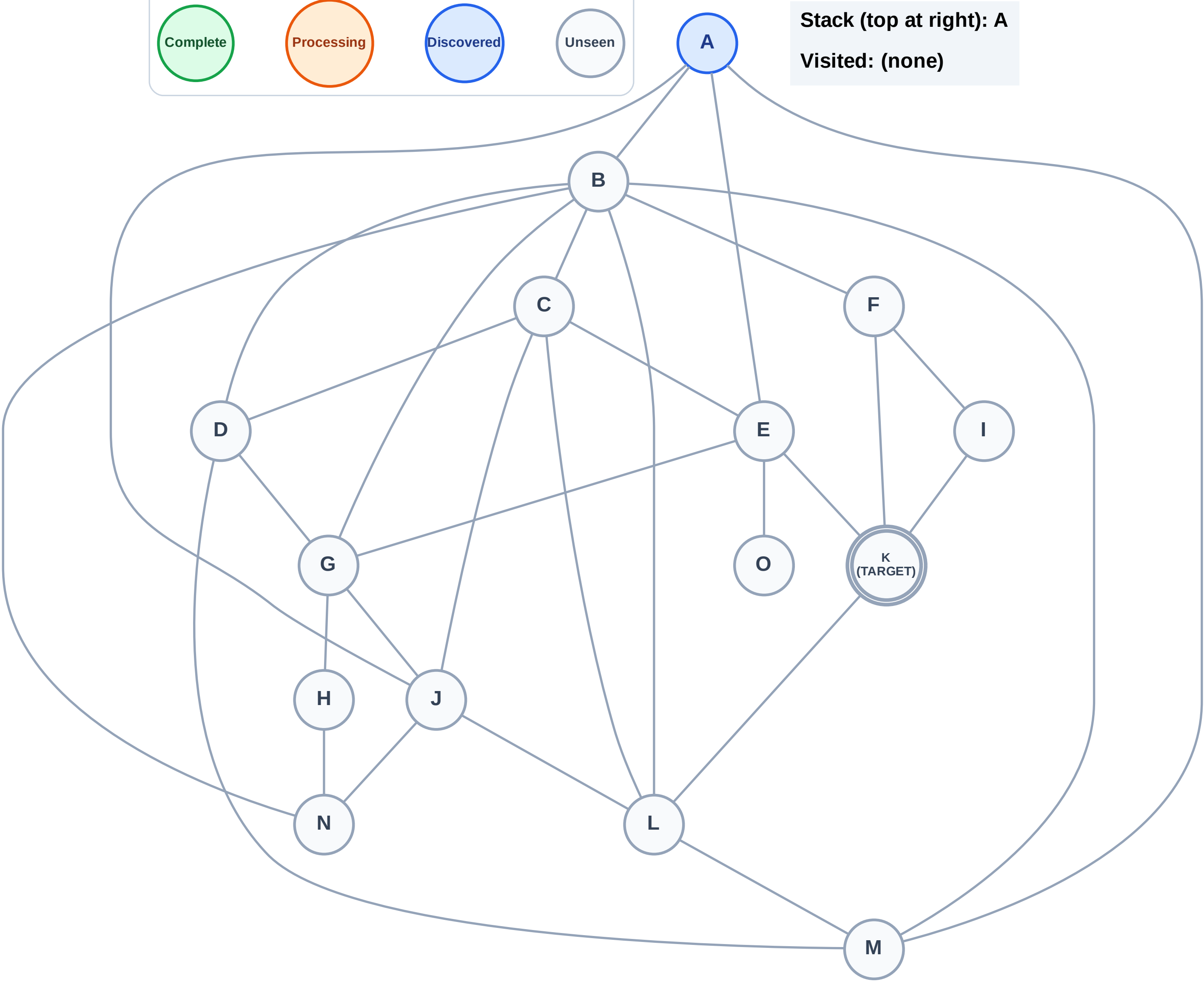
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

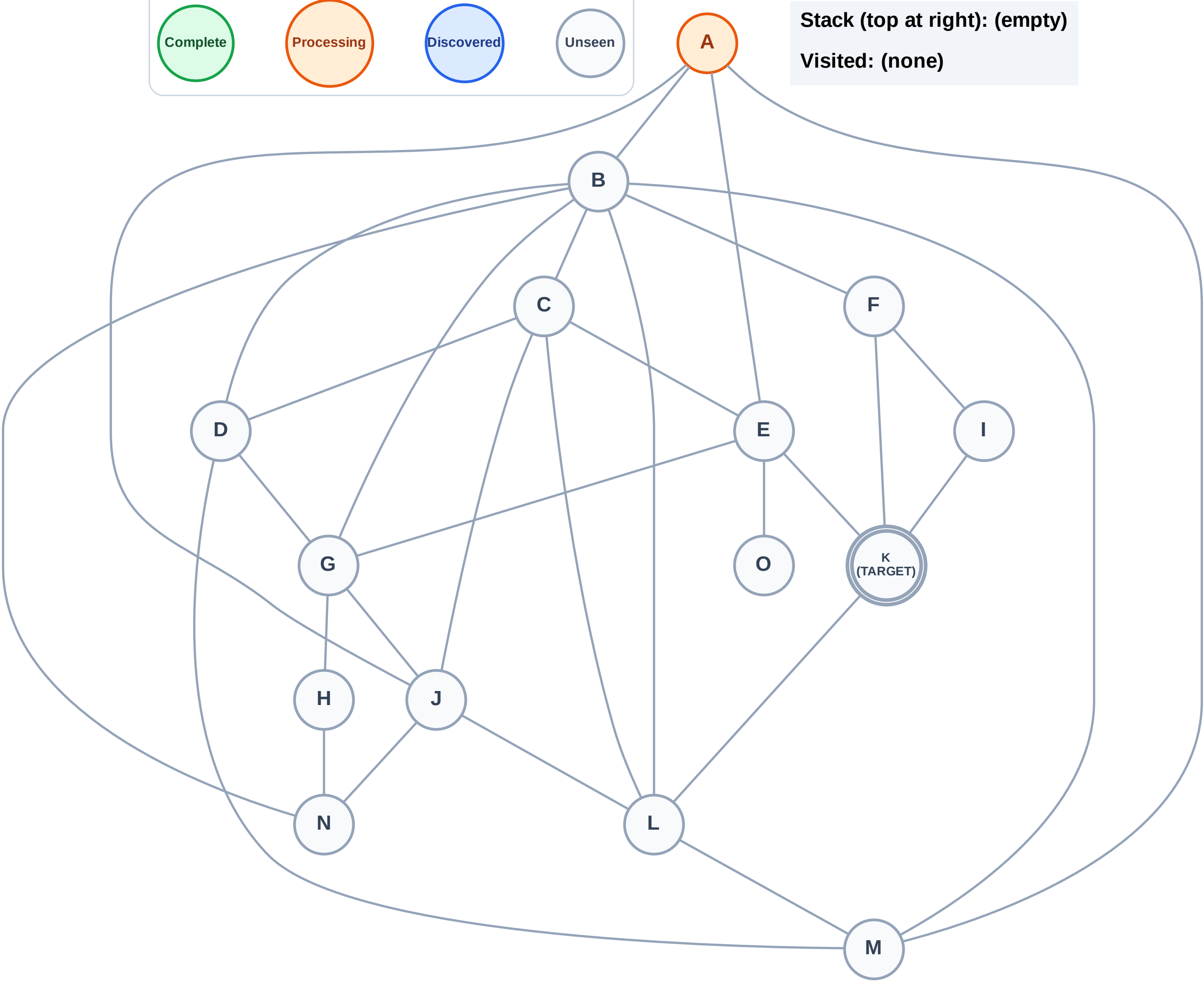
Complete

Processing

Discovered

Unseen

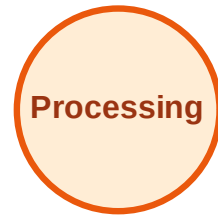
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

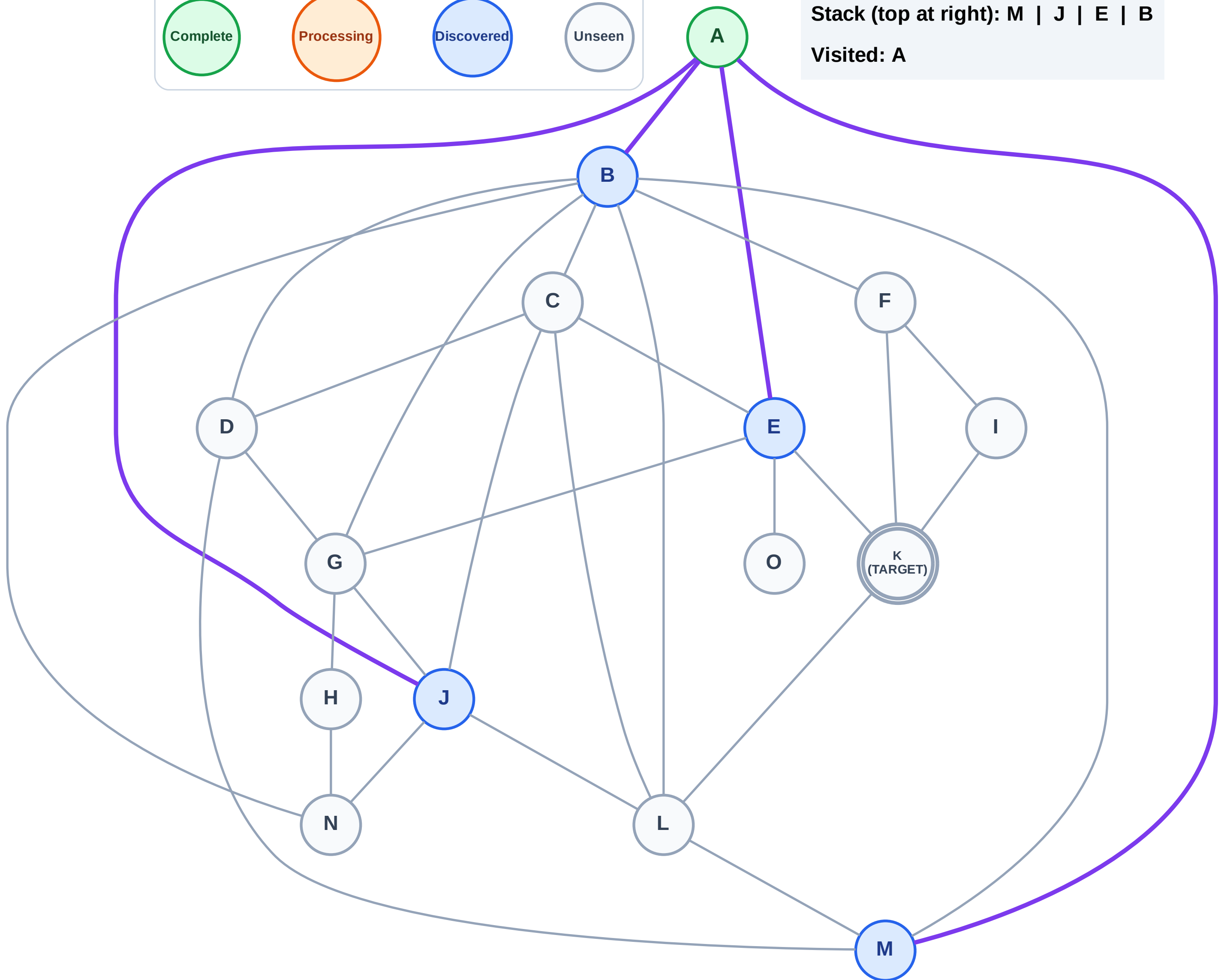
Step 3: Discover M, J, E, B from A

Legend



Stack (top at right): M | J | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

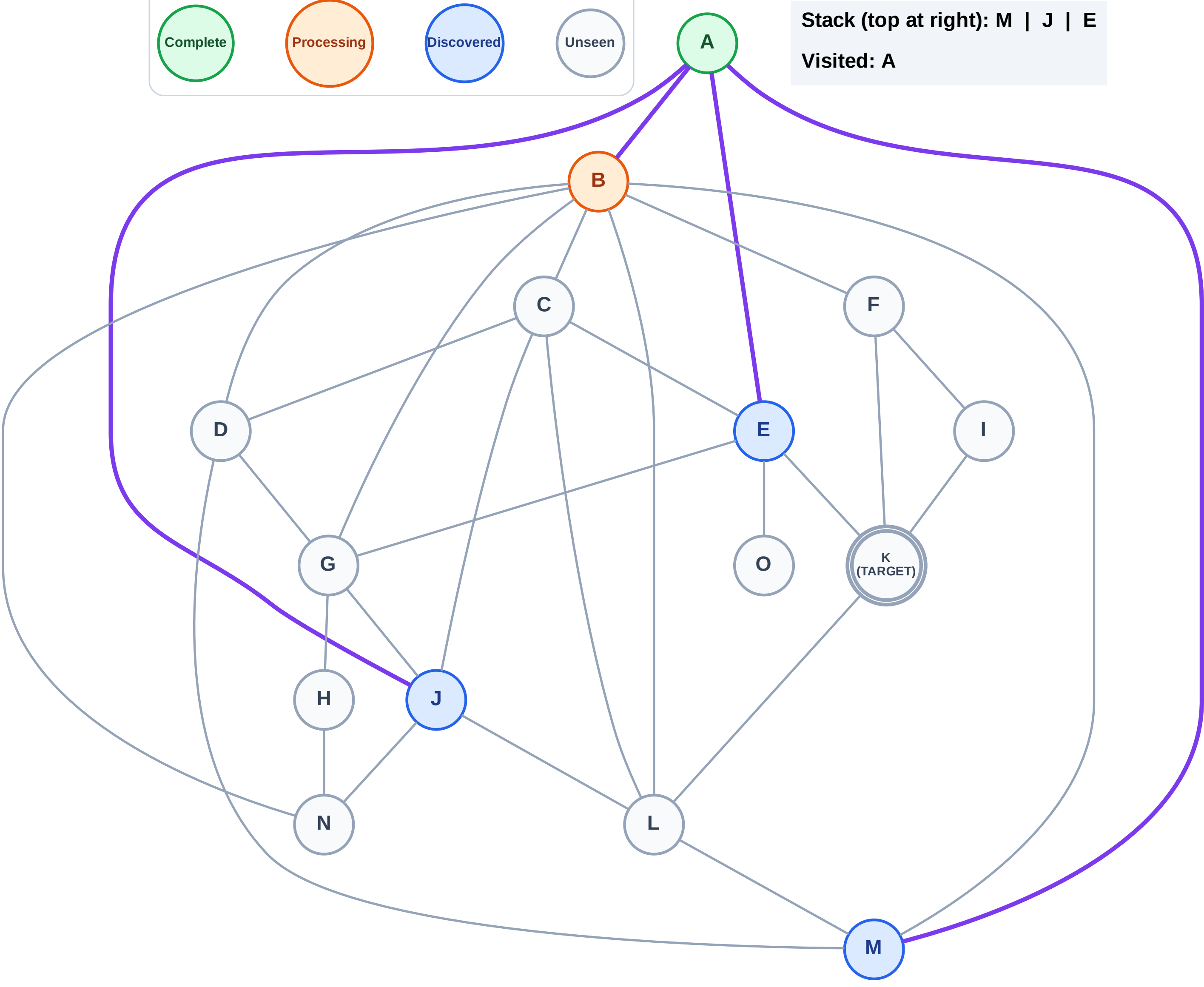
Processing

Discovered

Unseen

Stack (top at right): M | J | E

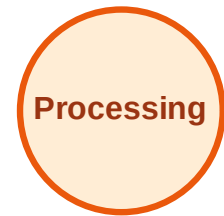
Visited: A



DFS Traversal

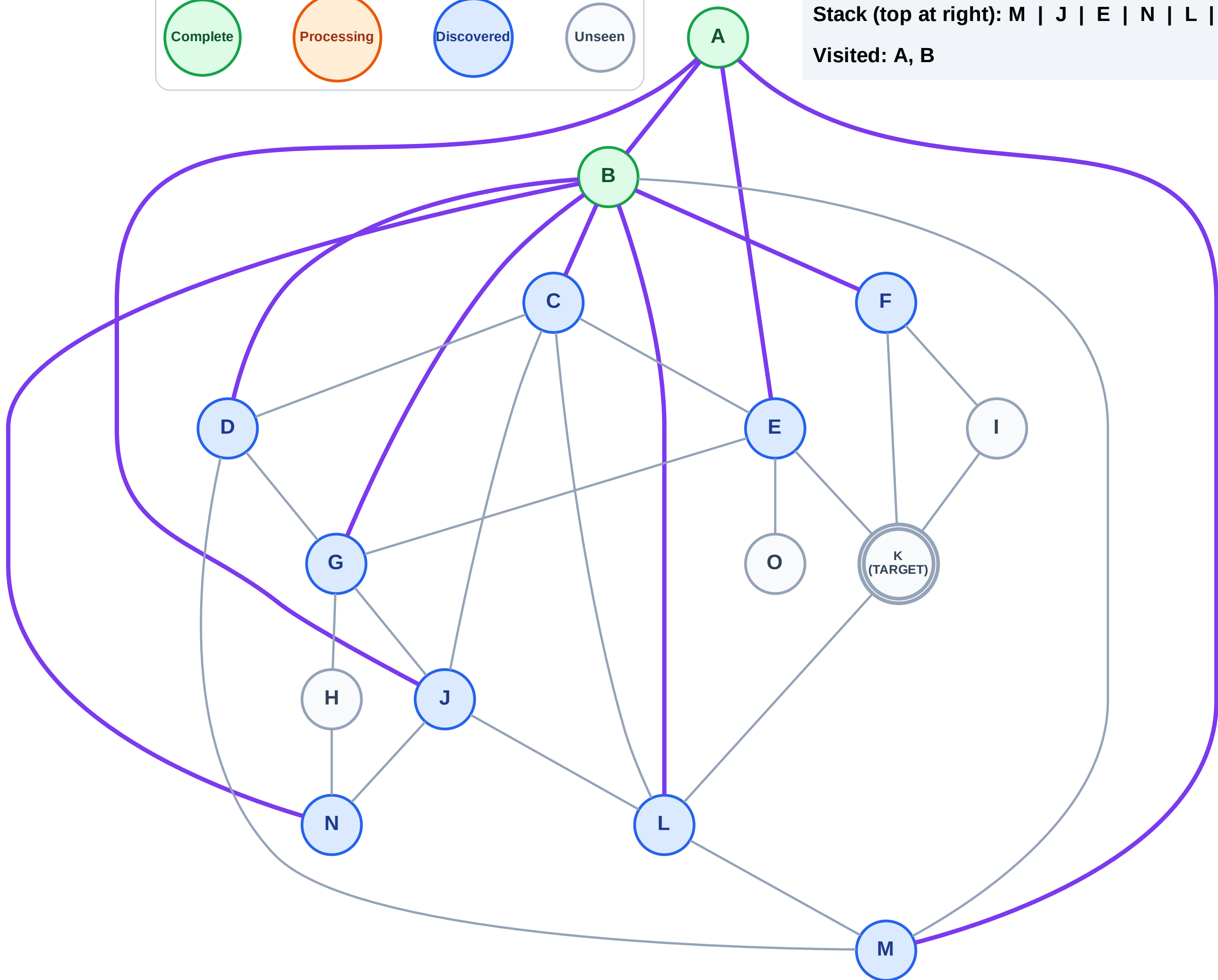
Step 5: Discover N, L, G, F, D, C from B

Legend



Stack (top at right): M | J | E | N | L | G | F | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

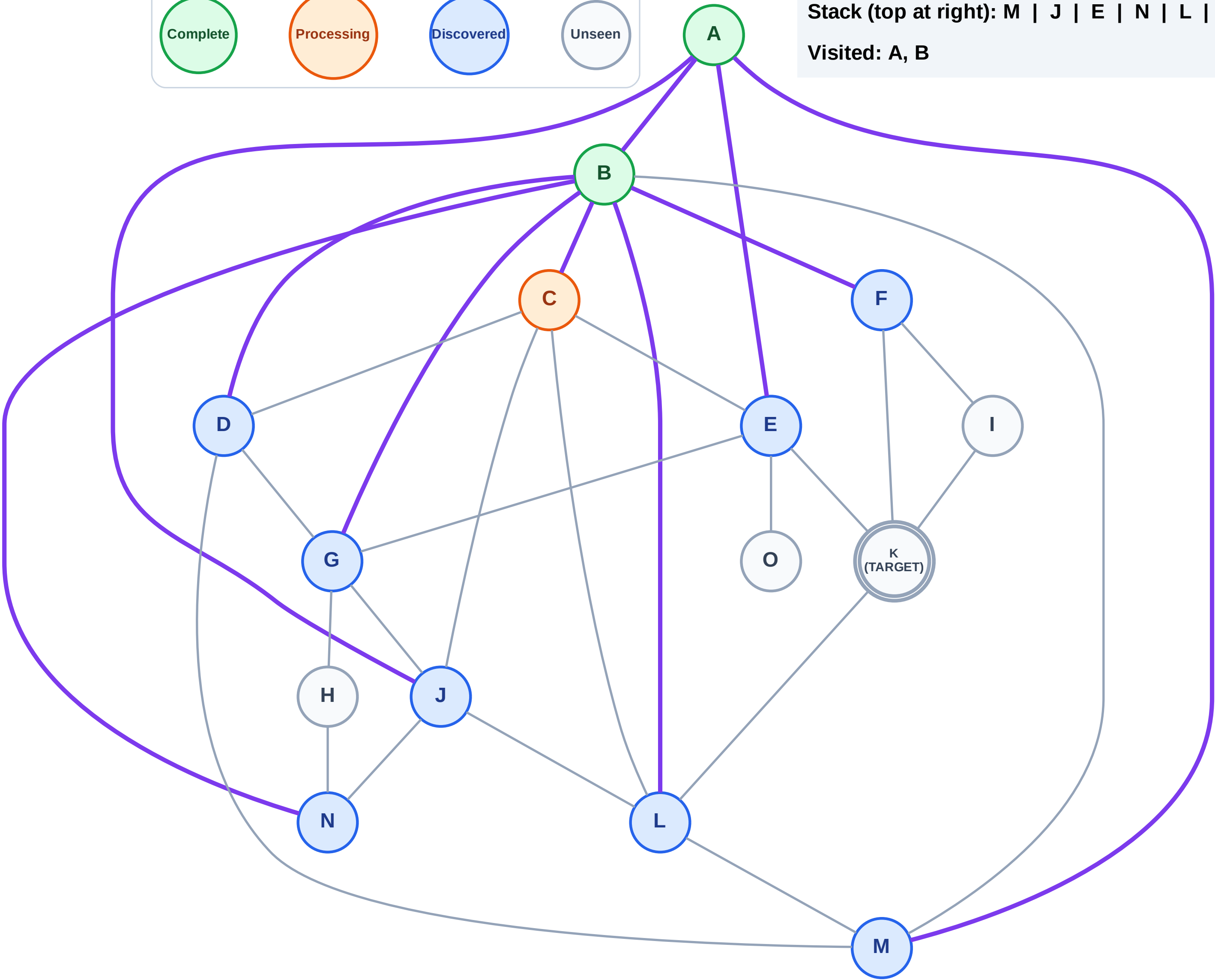
Processing

Discovered

Unseen

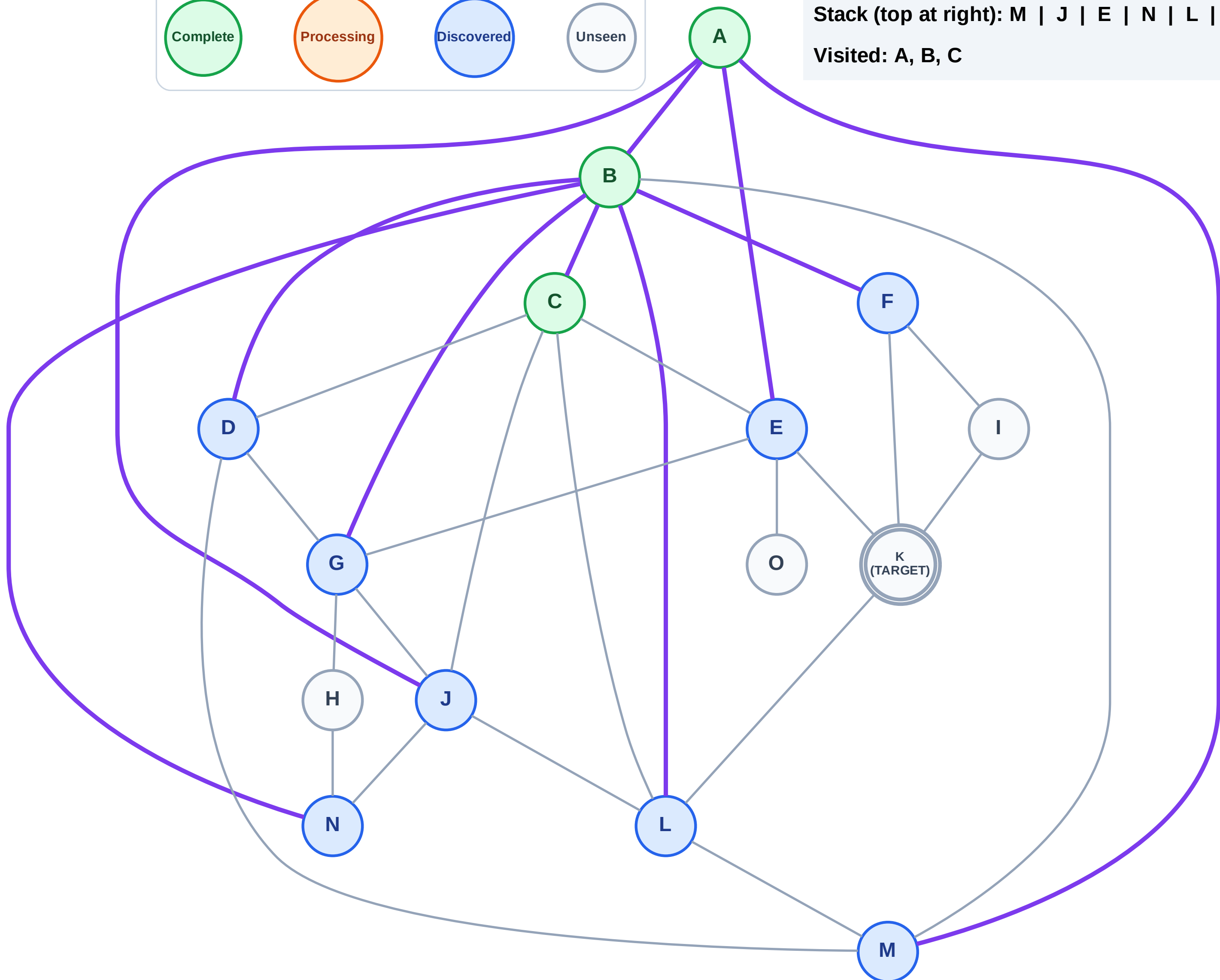
Stack (top at right): M | J | E | N | L | G | F | D

Visited: A, B



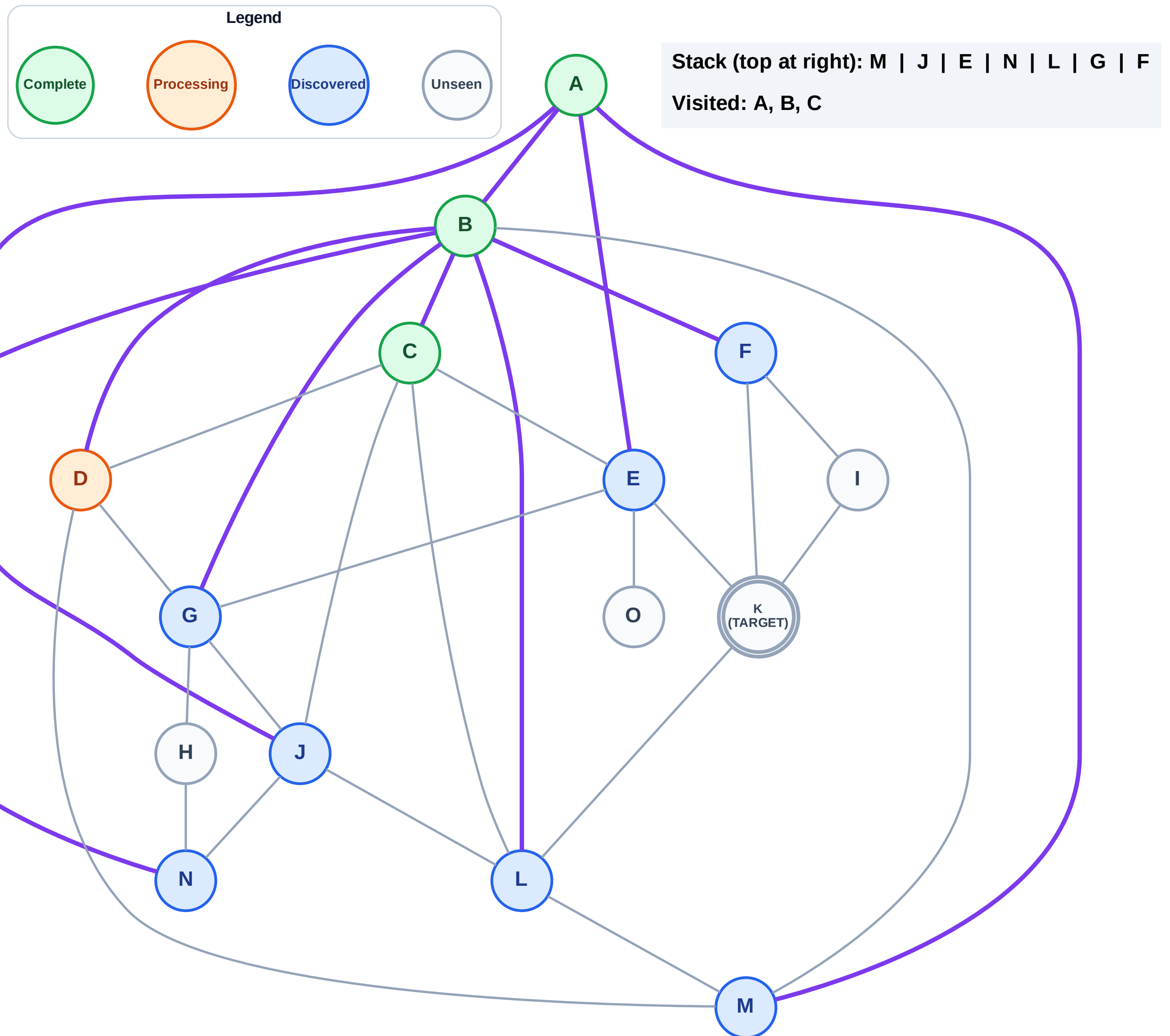
Step 7: No new neighbors from C

Step 7: No new neighbors from C



Step 8: Process node D

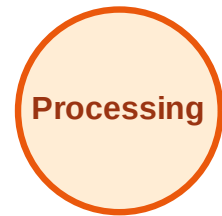
Step 8: Process node D



DFS Traversal

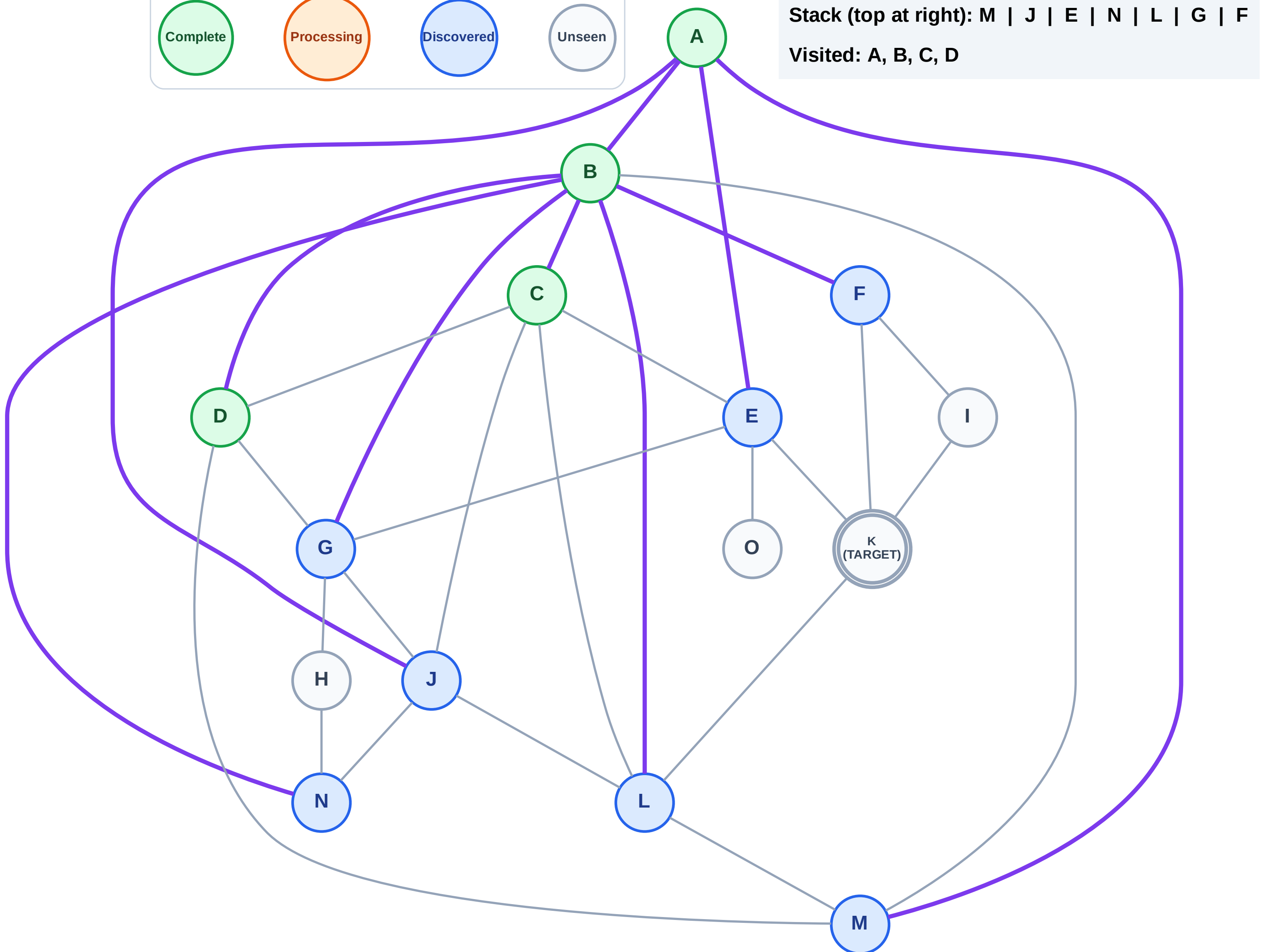
Step 9: No new neighbors from D

Legend



Stack (top at right): M | J | E | N | L | G | F

Visited: A, B, C, D



DFS Traversal

Step 10: Process node F

Legend

Complete

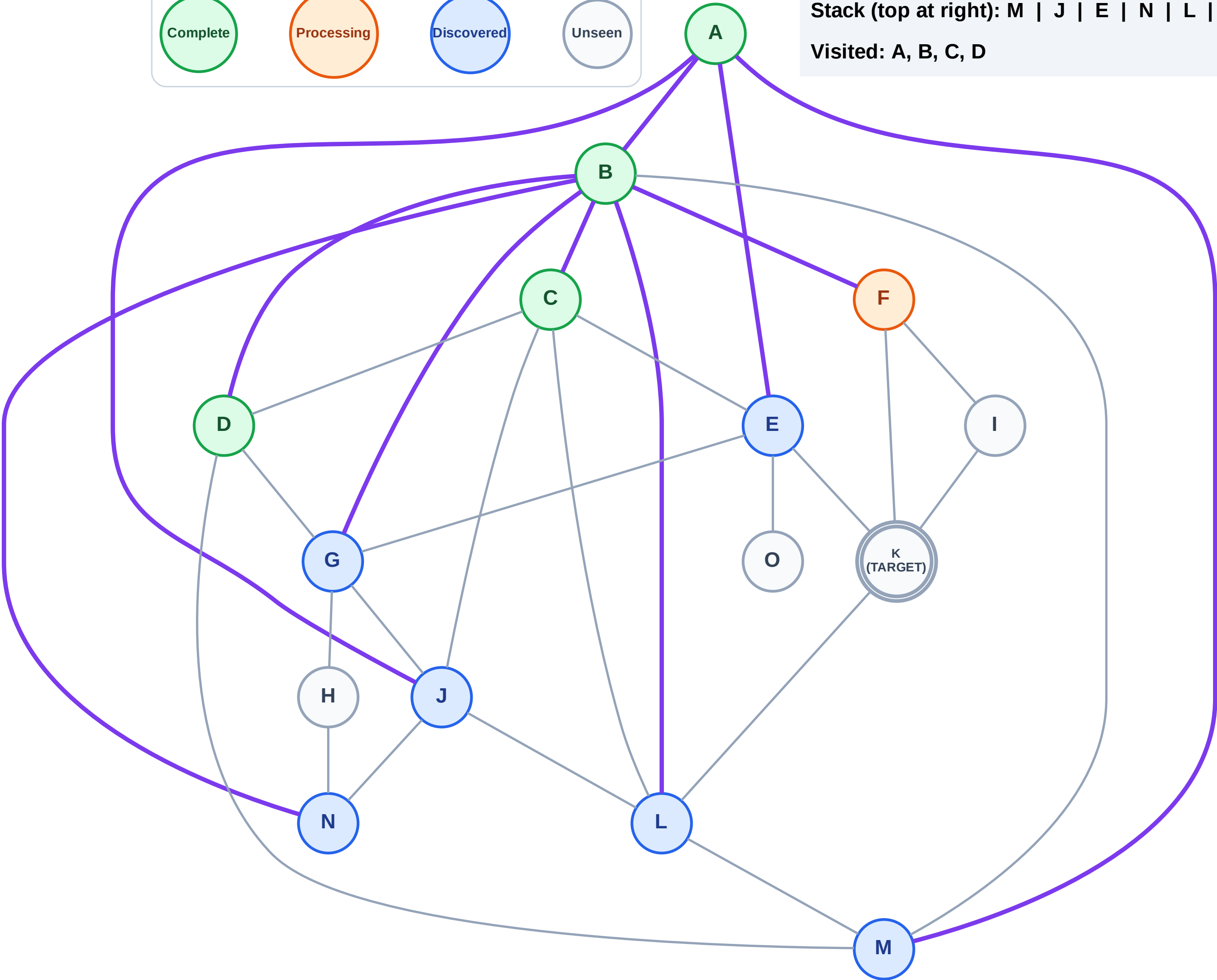
Processing

Discovered

Unseen

Stack (top at right): M | J | E | N | L | G

Visited: A, B, C, D



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | E | N | L | G

Visited: A, B, C, D, F, K

